

Model cards and parameter provenance

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Table of contents

1	Overview	1
2	Setup	2
3	A card for a model	2
4	Where the numbers came from	3
5	Validation, honestly	5
6	The report	5
7	Carrying the card with the model	6
8	Summary	7
9	References	7

1 Overview

Reviews of ecological belief-network practice keep reaching the same verdict: the networks are intelligible, but the reporting around them is not. Marcot et al. (2006) set out what a defensible model should say about itself; Chen and Pollino (2012) turned that into a practical checklist for environmental models; Kaikkonen et al. (2021), after screening several hundred studies, found elicitation methods and validation still reported too thinly to review or reuse. The gap is not an algorithm. It is that a model usually travels as a file of tables, with the decision it was built for, the meaning of its states, the source of each table and the limits of its use left in a paper, an inbox or somebody's memory.

A ModelCard is that missing record, held beside the model and serialised with it. This vignette builds one for the reference habitat network, records where each conditional probability table came from, writes the card as Markdown, and reads it back out of the JSON envelope. Nothing

here changes a kernel: a card is metadata, and two models with different cards are still the same model.

2 Setup

```
using BayesianNetworks

m = reference_habitat_model()
variable_names(syntax(m))
```

```
7-element Vector{Symbol}:
 :Climate
 :Irrigation
 :SoilMoisture
 :GrazingPressure
 :Vegetation
 :HabitatQuality
 :Occupancy
```

3 A card for a model

`ModelCard(m)` fills in everything that can be read off the model – the variables and their states, the mechanisms and the references they resolve to, and the history of any rewrites – and leaves the narrative fields for the modeller. Those fields are the ones a reviewer needs: which decision the model serves, which endpoint it predicts, over which spatial and temporal extent, why the graph looks the way it does, and which other structures were considered and rejected.

```
card = ModelCard(m;
    name = :habitat_occupancy,
    model_version = "1.0",
    decision_context = """
        Whether to reduce stocking rates on a 400 km2 reserve, and
where.
        """,
    endpoint = "Occupancy of the focal species at the end of the
season",
    spatial_extent = "the reserve, in 1 km grid cells",
    temporal_extent = "one growing season, 2026",
    graph_rationale = """
        Elicited at a two-day workshop: climate and irrigation drive
soil
        moisture, soil moisture and grazing drive vegetation, and
occupancy
        responds to habitat quality rather than to vegetation directly.
        """,
```

```

HabitatQuality",
    alternative_structures = ["Vegetation -> Occupancy without
                              "GrazingPressure -> HabitatQuality
                              directly"],
    elicitation_protocol = ""
    Four ecologists, three-point elicitation of each row, two
rounds
    with feedback, linear pooling of the second round.
    "",
    validation_summary = ""
    Temporal holdout: fitted on the 2019-2022 transects and scored
on
    2023-2024, against a baseline of the marginal occupancy rate.
    "",
    intended_use = "comparing stocking scenarios at reserve scale",
    limitations = ""
    Conditional scenario predictions, not evidence that a stocking
    change causes the predicted occupancy; not transferable off the
    reserve; no independent validation data yet.
    "",
    license = "CC BY 4.0")

```

ModelCard(:habitat_occupancy v1.0, 7 variables, 0/7 mechanisms with provenance)

The structural half is already populated, and per-variable state definitions start empty, which is deliberate: an empty field is a gap the report will show.

```
(card.variables, card.states[:HabitatQuality], length(card.mechanisms))
```

```
([:Climate, :Irrigation, :SoilMoisture, :GrazingPressure, :Vegetation,
:HabitatQuality, :Occupancy], [:poor, :good], 7)
```

State definitions say what a state *means* in the field, which is what makes a model reusable by someone who was not at the workshop. They are ordinary text, keyed by variable.

```

card.state_definitions[:Occupancy] = "detected in at least one of four surveys"
card.state_definitions[:Vegetation] = "mean sward height: <5 cm, 5-15 cm, >15 cm"
card.state_definitions[:HabitatQuality] = "composite index below or above 0.6"
sort(collect(keys(filter(p -> !isempty(last(p)), card.state_definitions))))

```

```

3-element Vector{Symbol}:
 :HabitatQuality
 :Occupancy
 :Vegetation

```

4 Where the numbers came from

Ecological networks mix estimates from data, numbers taken from the literature, values derived from process models and numbers elicited from experts, and the four are not equally strong

evidence ([Chen and Pollino 2012](#)). `ParameterProvenance` records, per mechanism, which kind a table is and everything needed to trace it: a citation, the dataset, the estimator, the expert or panel, a timestamp and free notes (the field list of section 49 of the package specification).

```
provenance!(card,
  [:SoilMoisture_mechanism =>
    ParameterProvenance(; source_type = :mechanistic,
      citation = "reserve water-balance model v3",
      estimator = "discretised model output",
      notes = "run on 2015-2025 rainfall"),
    :Vegetation_mechanism =>
    ParameterProvenance(; source_type = :empirical,
      dataset = "grazing transects 2019-2024, n = 812",
      estimator = "Dirichlet-multinomial, weak prior"),
    :HabitatQuality_mechanism =>
    ParameterProvenance(; source_type = :literature,
      citation = "Marcot et al. (2006), Table 2"),
    :Occupancy_mechanism =>
    ParameterProvenance(; source_type = :elicited,
      expert = "panel of four reserve ecologists",
      notes = "three-point elicitation, second round"))
provenance(card, :Vegetation_mechanism)
```

```
ParameterProvenance(:empirical, "", "grazing transects 2019-2024, n = 812",
  "Dirichlet-multinomial, weak prior", "", nothing, "")
```

The point of recording sources per mechanism is that the gaps become visible. `undocumented_mechanisms` lists the tables nobody has yet accounted for; here they are the three root variables, whose priors were taken from the file the model was read from.

```
undocumented_mechanisms(card)
```

```
3-element Vector{Symbol}:
 :Climate_mechanism
 :Irrigation_mechanism
 :GrazingPressure_mechanism
```

Asking for a mechanism with nothing recorded returns an `:unknown` provenance rather than an error, and a mechanism that is not the model's is refused, so a card cannot drift away from the network it describes.

```
err = try
  provenance!(card, :Rainfall_mechanism => ParameterProvenance())
catch e
  e
end
(provenance(card, :Climate_mechanism).source_type, sprint(showerror, err))
```

```
(:unknown, "UnknownMechanismError: no mechanism named :Rainfall_mechanism")
```

5 Validation, honestly

The card separates what was done to test the model from the scores that testing produced. The free-text summary, filled in above, is where the design of the test belongs – which data were held out, how, and against which baseline. The `validation_scores` slot holds the numbers, and is what `BayesianNetworkInference.jl`'s scoring layer fills; leaving it empty is itself a statement, which the report prints as one.

```
card.validation_scores[:brier] = 0.21
card.validation_scores[:log_score] = -0.65
card
```

```
ModelCard(:habitat_occupancy v1.0, 7 variables, 4/7 mechanisms with provenance, 2
scores)
```

6 The report

`report` writes the card as Markdown, in the order a reader needs it: context first, then the structure, then the numbers and their sources, then how the model was tested and what it must not be used for. Fields that were left empty are printed as *Not documented.*, so the report shows its own gaps rather than hiding them.

```
md = report(card)
println(join(first(split(md, "\n"), 24), "\n"))
```

```
# Model card: habitat_occupancy
```

```
## Decision context
```

```
Whether to reduce stocking rates on a 400 km2 reserve, and where.
```

```
## Endpoint and extent
```

- Endpoint: Occupancy of the focal species at the end of the season
- Spatial extent: the reserve, in 1 km grid cells
- Temporal extent: one growing season, 2026

```
## Graph rationale and alternatives
```

```
Elicited at a two-day workshop: climate and irrigation drive soil
moisture, soil moisture and grazing drive vegetation, and occupancy
responds to habitat quality rather than to vegetation directly.
```

- Vegetation -> Occupancy without HabitatQuality
- GrazingPressure -> HabitatQuality directly

```
## Variables and states
```

The provenance table is the part that a reviewer will read first.

```
sections = split(md, "## ")
println("## ", first(filter(s -> startswith(s, "Mechanisms"), sections)))
```

```
## Mechanisms and parameter provenance
```

Mechanism	Reference	Source	Citation	Dataset	Estimator	Expert	Timestamp	Notes
SoilMoisture_mechanism	SoilMoisture_mechanism	mechanistic	reserve	water-balance model v3	discretised model output	run on 2015-2025	rainfall	
Vegetation_mechanism	Vegetation_mechanism	empirical	grazing transects	2019-2024, n = 812	Dirichlet-multinomial, weak prior			
HabitatQuality_mechanism	HabitatQuality_mechanism	literature	Marcot et al. (2006), Table 2					
Occupancy_mechanism	Occupancy_mechanism	elicited	panel of four reserve ecologists	three-point elicitation, second round				
Climate_mechanism	Climate_mechanism	*not documented*						
Irrigation_mechanism	Irrigation_mechanism	*not documented*						
GrazingPressure_mechanism	GrazingPressure_mechanism	*not documented*						

Mechanisms without provenance: `Climate\_mechanism`, `Irrigation\_mechanism`, `GrazingPressure\_mechanism`.

7 Carrying the card with the model

A card that lives in a separate document is a card that gets lost. `json_model` takes one and writes it into the same envelope as the syntax, semantics, evidence and history; `read_json_card` reads it back, and returns nothing for a file written without one, so older model files still read exactly as before.

```
path = tempname()
write_json_model(path, m; card = card)
(read_json_card(path) == card, read_json_model(path) == m)
```

```
(true, true)
```

```
plain = tempname()
write_json_model(plain, m)
(read_json_card(plain), read_json_model(plain) == m)
```

```
(nothing, true)
```

Because a card is metadata, it never enters the semantics: the same model with and without a card is the same value, evaluates identically, and compares equal.

```
(read_json_model(path) == read_json_model(plain),  
 marginal(read_json_model(path), :Occupancy) ≈ marginal(m, :Occupancy))
```

```
(true, true)
```

```
rm(path)  
rm(plain)
```

8 Summary

- A `ModelCard` records what the ecology reporting standard asks of a model: decision context, endpoint and extent, graph rationale and the alternatives considered, state definitions, parameter provenance, elicitation protocol, validation, intended use, limitations, licence and version.
- `ModelCard(m)` prefills the structural half from a model; the narrative fields are the modeller's to write, and empty ones are reported as gaps rather than omitted.
- `ParameterProvenance` records per mechanism whether a table is empirical, elicited, from the literature or mechanistic, with the citation, dataset, estimator, expert and timestamp needed to trace it; `undocumented_mechanisms` lists what is still missing.
- `report` writes the card as Markdown, and `json_model(m; card)` carries it in the model's own JSON envelope, with files that predate cards reading unchanged.
- A card is metadata: it never changes a kernel, an evaluation or model equality.

This is the last vignette in the series; the scoring layer that fills `validation_scores` lives in `BayesianNetworkInference.jl`.

9 References

- Chen, Serena H., and Carmel A. Pollino. 2012. "Good Practice in Bayesian Network Modelling." *Environmental Modelling & Software* 37: 134–45. <https://doi.org/10.1016/j.envsoft.2012.03.016>.
- Kaikkonen, Laura, Tuuli Parviainen, Mika Rahikainen, Laura Uusitalo, and Annukka Lehtikoinen. 2021. "Bayesian Networks in Environmental Risk Assessment: A Review." *Integrated Environmental Assessment and Management* 17 (1): 62–78. <https://doi.org/10.1002/ieam.4332>.

Marcot, Bruce G., J. Douglas Steventon, Glenn D. Sutherland, and Robert K. McCann. 2006. "Guidelines for Developing and Updating Bayesian Belief Networks Applied to Ecological Modeling and Conservation." *Canadian Journal of Forest Research* 36 (12): 3063–74. <https://doi.org/10.1139/x06-135>.